Contents

Chapter 1

Results and Discussion

Following the experimental and analysis procedures explained in Chapter 2, the temperature of the atomic cloud and the ODT capture efficiency were characterized and optimized, and the stability of the optical field delivered through the black fiber was assessed. This chapter is organized in four parts.

The first part addresses the optimization of the molasses temperature, carried out by adjusting the cooling-laser detuning, the cooling-laser intensity, and the compensation magnetic field (Sec. 1.1).

The second part describes the optimization of the ODT loading. The compensation coils are used to displace the MOT after loading, in order to maximize the spatial overlap between the atomic cloud and the ODT beam and thereby enhance the capture efficiency; the measured loading efficiency is then compared with the capture model of Chapter 2 (Sec. 1.2).

The third part presents the characterization of the power and polarization stability of the field transmitted through the black fiber, analyzed in the time domain with the Allan deviation and in the frequency domain with the amplitude spectral density, and discusses the implications of the measured noise for the optical conveyor belt (Sec. 1.3).

The fourth part reports the numerical simulation of EIT cooling, starting from the ideal three-level system and extending to the full 24-level structure of the ^{87}Rb D_2 line, including the effect of optical pumping and repumping on the cooling dynamics (Sec. 1.4).

1.1 Molasses temperature optimization

Prior to the optimization procedure, the six cooling beams were realigned in order to minimize their spatial mismatch at the MOT position, as thermal drifts may gradually alter the alignment of the optical components.

Before realignment, preliminary values of the compensation coil currents had been determined:

$$\begin{cases} I_{\mathrm{x}} & = 80 \text{ mA} \quad (B_x \approx 11.2\ \mu\mathrm{T}), \\ I_{\mathrm{y}} & = -200 \text{ mA} \quad (B_y \approx -26.1\ \mu\mathrm{T}), \\ I_{\mathrm{z}} & = -400 \text{ mA} \quad (B_z \approx -60.0\ \mu\mathrm{T}). \end{cases} \tag{1.1}$$

These values served as the initial settings for the optimization described below. Starting from an initial value of approximately 20 μK, the optimization procedure minimized the optical molasses temperature. Under the final conditions, a minimum average temperature of $T = (8.42 \pm 0.17)$ μK was achieved.

The molasses temperature was determined from the ballistic expansion of the atomic cloud. Figure 1.1 presents fluorescence images of the freely falling atomic cloud recorded at different TOFs under the optimized conditions. The cloud widths extracted from Gaussian fits were then fitted according to equation (2.14), as shown in Figure 1.2.

The temperatures extracted along the two Cartesian directions are $T_x = (7.92 \pm 0.20)$ μK and $T_y = (8.91 \pm 0.28)$ μK. The slight anisotropy observed between the two axes is likely due to small imperfections in the cooling beams' alignment or intensity balances. Their mean gives the final average temperature of $T = (8.42 \pm 0.17)$ μK.

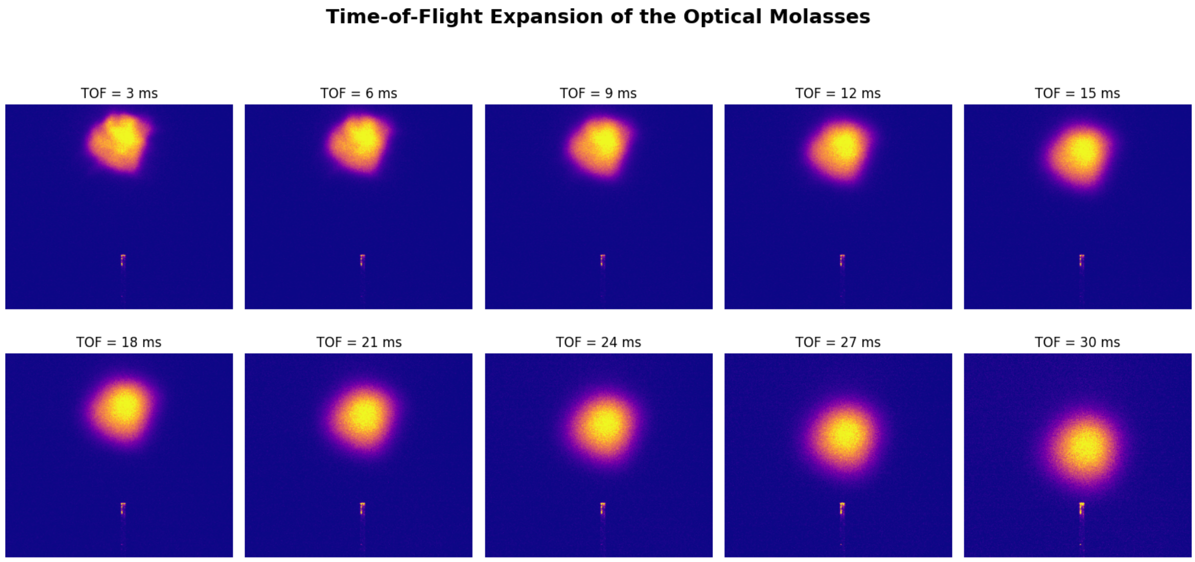


Figure 1.1: *Fluorescence images of the atomic cloud recorded during Time-of-Flight expansion, with expansion times ranging from 3.0 ms to 30.0 ms.*

As shown in Figure 1.2, the measurements at long TOF deviate from the linear trend and lie above the fitted values. The same behavior was observed systematically throughout the measurement series. A plausible explanation is that, after sufficiently long expansion times, the cloud radius is no longer negligible compared with the probe beam radius, so that the assumption of uniform illumination is no longer valid. Atoms in the outer region of the cloud are therefore exposed to a weaker probe intensity, this effect should become less relevant as the atomic temperature decreases.

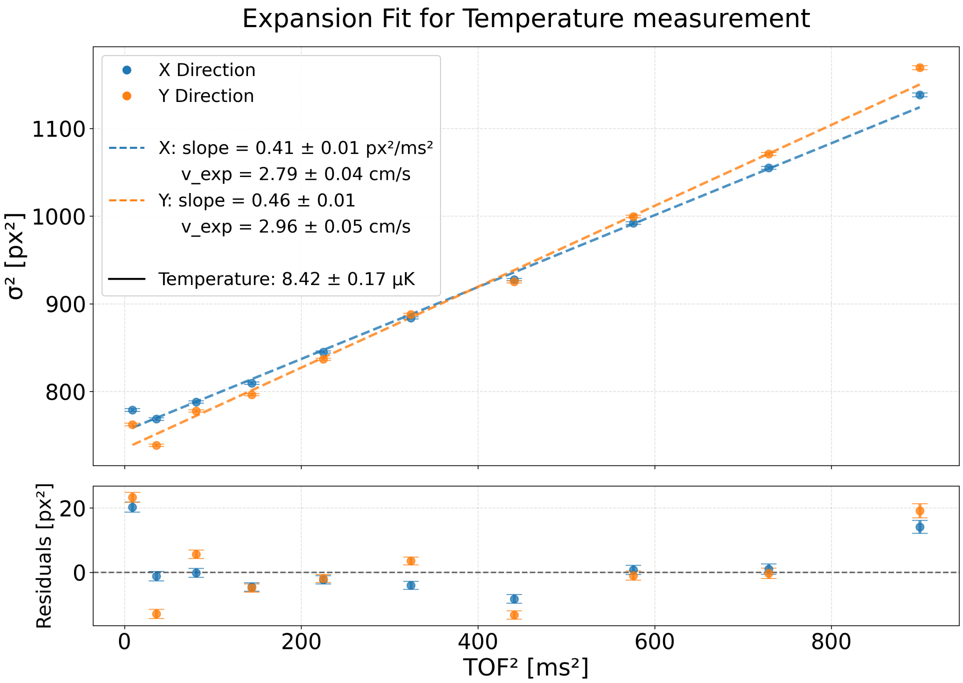


Figure 1.2: *Temperature fit extracted from the time-of-flight (TOF) expansion of the optical molasses, displayed alongside the corresponding residuals.*

1.1.1 Detuning frequency optimization

Starting from an initial molasses temperature of approximately 20 μK, the cooling laser detuning was optimized in order to reach the final temperature reported above. The detuning was scanned from -24.4 MHz ($4.0\ \Gamma/2\pi$) to -184 MHz ($30.4\ \Gamma/2\pi$) in steps of -12.1 MHz ($2.0\ \Gamma/2\pi$).

Figure 1.3 shows the measured average temperature as a function of the absolute detuning, expressed in units of $\Gamma/2\pi$. For relatively small detunings, the measurements exhibit the $1/|\delta|$ scaling expected in the sub-Doppler cooling regime. At larger detunings, two additional effects become important: the optical scattering rate decreases rapidly as $\Gamma_p \propto 1/\delta^2$, while the cooling frequency approaches the $F = 2 \to F' = 2$ transition, which lies 267 MHz below the $F = 2 \to F' = 3$ cooling transition (see Figure 2.2). The combined effect causes the temperature curve to level off around $|\delta| \approx 15 - 20$ and to increase again at still larger detunings.

The minimum average temperature was measured at a normalized detuning of $|\delta| = 17.2$, corresponding to an absolute frequency detuning of 104.2 MHz. The detuning optimization therefore reduced the molasses temperature from its initial value of approximately 20 μK to the final value of $T = (14.9 \pm 0.4)\ \mu$K.

To quantitatively compare the sub-Doppler cooling performance with existing literature,

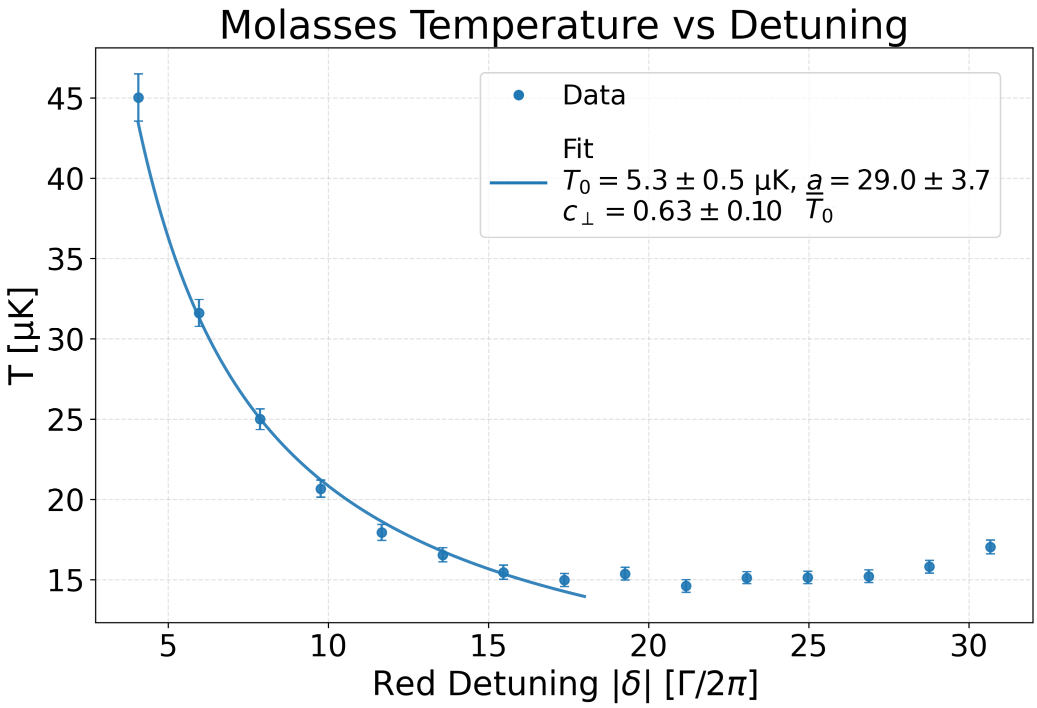


Figure 1.3: *Optical molasses temperature as a function of the cooling laser detuning. The horizontal axis shows the detuning* $|\delta|$*, normalized to the natural linewidth* $\Gamma/2\pi$*. The most effective sub-Doppler cooling is observed in the interval* $|\delta| \approx 15 - 20$*, where a minimum temperature of* 14.9 ± 0.4 μK *is reached. Up to approximately* $|\delta| = 18$*, the temperature follows the* $1/|\delta|$ *scaling predicted for ideal Sisyphus cooling.*

a fit was performed up to $|\delta| = 18$. The model fitted to the data is

$$T = a\frac{\Gamma}{|\delta|} + T_0. \tag{1.2}$$

Following the framework by Gerz *et al.* [1], the temperature of a 3D optical molasses scales linearly with the AC Stark shift as

$$\frac{k_B T}{\hbar\Gamma} = c_\perp \frac{\Omega^2}{|\delta|\Gamma} + \text{const.} \tag{1.3}$$

The fit parameter a is thus related to the dimensionless constant $c_\perp$. By introducing the Doppler temperature $T_D = \hbar\Gamma/(2k_B)$ and the saturation parameter $s_0 = I_0/I_{\text{sat}}$, where the squared Rabi frequency relates to intensity as $\Omega^2/\Gamma^2 = s_0/2$, the explicit relation is obtained:

$$c_\perp = \frac{a}{s_0 T_D}. \tag{1.4}$$

For the transition used, the saturation intensity is $I_{\text{sat}} = 1.6$ mW/cm^2. Given a single cooling beam power of $P = 6.6$ mW and a waist radius of $w = 12.5$ mm, the single-beam peak intensity is evaluated as $I_0 = 2P/(\pi w^2) \approx 2.69$ mW/cm^2. This yields a saturation

parameter $s_0 \approx 1.68$.
The fit yields $a = 153 \pm 24$ μK and a temperature offset $T_0 = 5.3 \pm 0.5$ μK (Figure 1.3). Using $T_D \approx 146$ μK and the calculated value of s_0, the a parameter converts to $c_\perp = 0.63 \pm 0.10$.
This result can be compared with existing values for ^{87}Rb in a 3D lin $\perp$ lin optical molasses. Literature reports experimental $c_\perp$ values ranging from 0.37 to 0.41 [1], and a theoretical value of $c_\perp = 0.31$ when accounting for atomic localization effects. The measured value of $c_\perp = 0.63 \pm 0.10$ is higher than previously reported experimental ranges, but remains on the same order of magnitude, confirming the expected Sisyphus cooling dynamics.

1.1.2 AOM amplitude optimization

The second stage of the optimization focused on the effect of the cooling beam intensity. This was investigated by varying both the AOM drive amplitude and the duration of the amplitude ramp applied at the beginning of the molasses phase. The AOM amplitude, given in arbitrary units (a.u.), determines the RF amplitude used to drive the modulator. Three settings were examined: 385, 475, and 550 a.u., corresponding respectively to cooling laser powers of 5.8 mW, 6.6 mW, and 6.3 mW measured after the collimator of the MOT z-beam. These yield single-beam peak intensities of 2.36 mW/cm^2, 2.69 mW/cm^2, and 2.57 mW/cm^2 (saturation parameters $s_0 \approx 1.48$, 1.68, and 1.60). The MOT itself is operated at an AOM amplitude of 550 a.u.
Ramp durations of 5 ms and 20 ms were also compared for the transition between the MOT and molasses amplitude settings. The measurements were repeated at several values of the cooling laser detuning.
The results are summarized in Figure 1.4. Although modest differences in the measured temperature are visible among the tested AOM amplitudes, they remain compatible within the experimental uncertainties. Consequently, no statistically meaningful dependence of the molasses temperature on the AOM amplitude could be established, and no unique optimal value was identified over the range considered.
By contrast, Figure 1.4 shows that the 20 ms ramp consistently produces lower temperatures than the 5 ms ramp, with an average reduction of approximately 0.5 μK. Since this improvement was modest but adequate for the experimental objectives, additional ramp durations were not explored. A ramp time of 20 ms was therefore adopted for the following measurements.
During the subsequent characterization of the ODT loading, however, this configuration was found to introduce additional fluctuations in the system. The compensation coil currents were therefore optimized using an AOM amplitude of 385 a.u., which yielded the lowest measured temperature at $|\delta| = 17.2$ $\Gamma/2\pi$, together with a ramp duration of

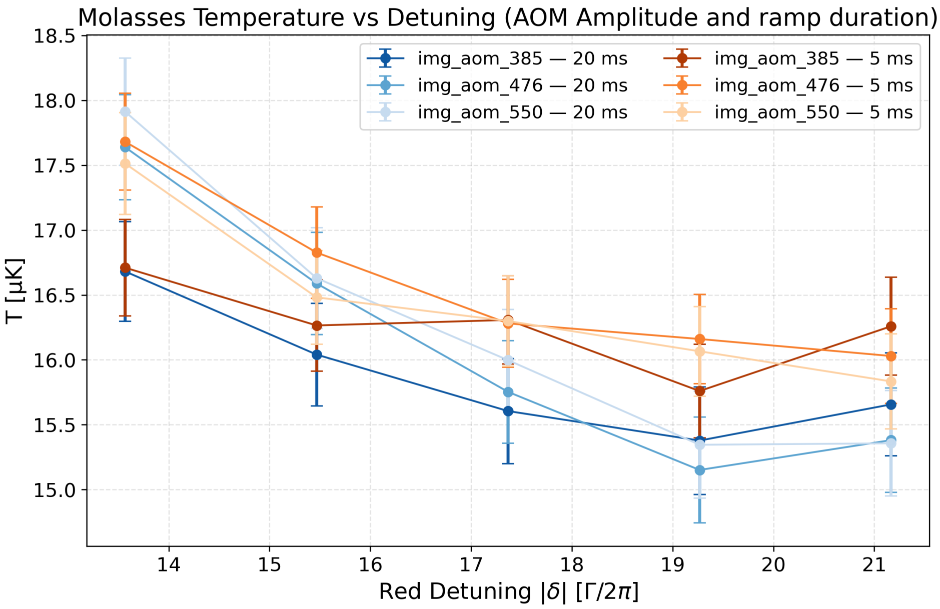


Figure 1.4: *Optical molasses temperature as a function of cooling laser detuning for different AOM amplitudes and ramp durations. No clear optimum in the AOM amplitude is visible, since the measured variations remain within the experimental uncertainties. In contrast, the temperatures obtained with a 20 ms ramp (blue lines) are systematically lower than those measured with a 5 ms ramp (red lines).*

20 ms. These settings were later modified during the optimization of the ODT loading. Nevertheless, the measurements discussed here indicate that changes in these parameters affect the molasses temperature only at the level of a few tenths of a microkelvin and therefore have no substantial impact on the minimum temperature ultimately achieved.

1.1.3 Compensation field optimization

The final stage of the molasses optimization consisted of adjusting the magnetic compensation field. The currents supplied to the compensation coils were switched to their molasses values 5 ms after the MOT magnetic field was turned off.

The three current components were optimized sequentially: the z-axis current was adjusted first, while the x- and y-axis currents were held at the preliminary values given in equation (1.1). The x-axis current was then optimized using the newly determined value of the z component. Finally, the y-axis current was adjusted while keeping both the x- and z-axis currents at their optimized settings. The experimental data were fitted with quadratic functions to determine the optimal currents and the associated minimum temperatures. The results for the z-, x-, and y-axis coils are presented in Figure 1.5(a), (b), and (c), respectively.

The optimized current settings subsequently used during ODT loading were:

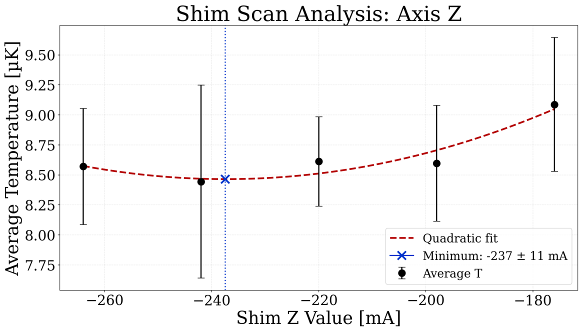
$$\begin{cases} I_x &= -15 \text{ mA}, \\ I_y &= -185 \text{ mA}, \\ I_z &= -230 \text{ mA}. \end{cases} \tag{1.5}$$

Using the calibration reported in Section 2.1.1, these currents correspond to the following magnetic field components:

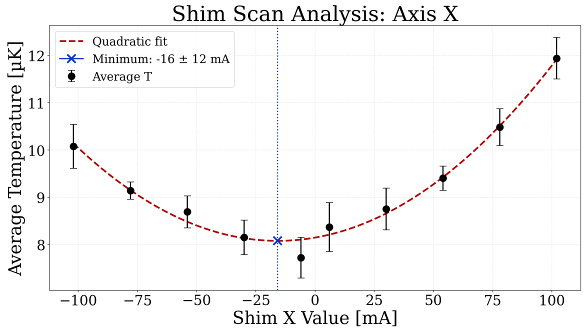
$$\begin{cases} B_x &= -21 \text{ mG}, \\ B_y &= -241 \text{ mG}, \\ B_z &= -345 \text{ mG}. \end{cases} \tag{1.6}$$

With these parameters, the molasses temperature was reduced to $T = (8.4 \pm 0.2)$ μK. The largest correction relative to the preliminary settings occurred along the z axis. A smaller, though still substantial, adjustment was required for the x-axis current, whereas the optimized y-axis current remained close to its initial value.

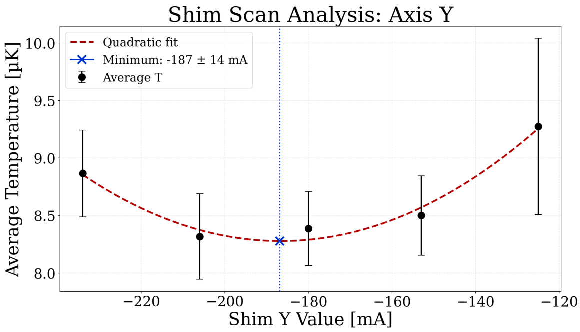
The operating values were chosen to be integers close to the fitted minima. Since in all three cases these settings lie near the vertex of the fitted parabola, the slight deviation from the exact optimal values is not expected to introduce a significant systematic contribution to the temperature measurements.



(a)



(b)



(c)

Figure 1.5: *Optical molasses temperature as a function of the compensation currents. (a) Shim-Z scan: the minimum is located at* $I_z = -237.5 \pm 11$ *mA. (b) Shim-X scan: the parabolic fit indicates a minimum at* $I_x = -16 \pm 12$ *mA. (c) Shim-Y scan: the minimum is found at* $I_y = -187 \pm 14$ *mA.*

1.2 ODT loading optimization

The molasses parameters determined in the previous section were used as the starting point for optimizing the loading of the optical dipole trap. This procedure aimed to maximize the spatial overlap between the cold atomic cloud and the ODT beam by adjusting the compensation coil currents.

The MOT is formed near the minimum of the magneto-optical potential. Applying uniform magnetic fields through the compensation coils displaces the magnetic-field zero and, consequently, the equilibrium position of the trapped cloud. The MOT can therefore be translated toward the optical dipole trap axis before the molasses phase begins. Although moving the cloud away from the optimal intersection of the cooling beams may cause moderate atom loss and heating, it can improve the overlap with the dipole-trap mode and increase the number of atoms loaded into the ODT.

After the initial MOT loading stage, the compensation coil currents were ramped to their ODT-loading values over 200 ms in 20 discrete steps. The gradual translation was introduced to limit heating and atom loss during the displacement. Since the required shift was small compared with the dimensions of the cooling beams, the optimized molasses parameters were retained throughout the ODT-loading measurements.

Preliminary compensation currents, determined before the optical realignment and molasses optimization, were used as the initial settings:

$$\begin{cases} I_x &= -200 \text{ mA}, \\ I_y &= -350 \text{ mA}, \\ I_z &= 50 \text{ mA}. \end{cases} \tag{1.7}$$

During these measurements, the ODT laser power before injection was 1.4 W. With a transmission efficiency of approximately 42%, the corresponding power was about 0.59 W. Since additional optical components are located upstream of the vacuum apparatus, this value provides a conservative estimate of the power at the fiber facet. A relative uncertainty of 10% was assigned, yielding

$$P = (0.59 \pm 0.06) \text{ W}. \tag{1.8}$$

1.2.1 Optimization of the detection conditions

Before scanning the compensation coil currents, the acquisition conditions were adjusted to ensure that the weak ODT signal could be extracted reliably from the fluorescence images. The preliminary measurements were performed with the FLIR camera using the currents in equation (1.7). The analysis was based on the differential-imaging procedure described in Section 2.2.5; all reported camera counts, integrated signal areas, and intensity values are expressed in arbitrary units [a.u.].

These initial measurements revealed two main limitations. First, the residual background after subtraction was comparable to, or larger than, the signal produced by the atoms captured in the ODT. Second, because of the small transverse size of the dipole trap, the ODT contribution extended over only a few pixels in the FLIR images. The combination of limited spatial sampling and background fluctuations prevented a robust quantitative determination of the trapped signal.

The background stability was found to depend on the AOM amplitude ramp applied during the molasses phase. Shortening the ramp duration from 20 ms to 5 ms improved the repeatability of the fluorescence profiles and reduced the residual background after subtraction. The final AOM amplitude was set to 480 a.u., corresponding to a cooling power of 6.6 mW measured on the MOT z-beam. This operating point lies near the saturation region of the AOM diffraction-efficiency curve, where

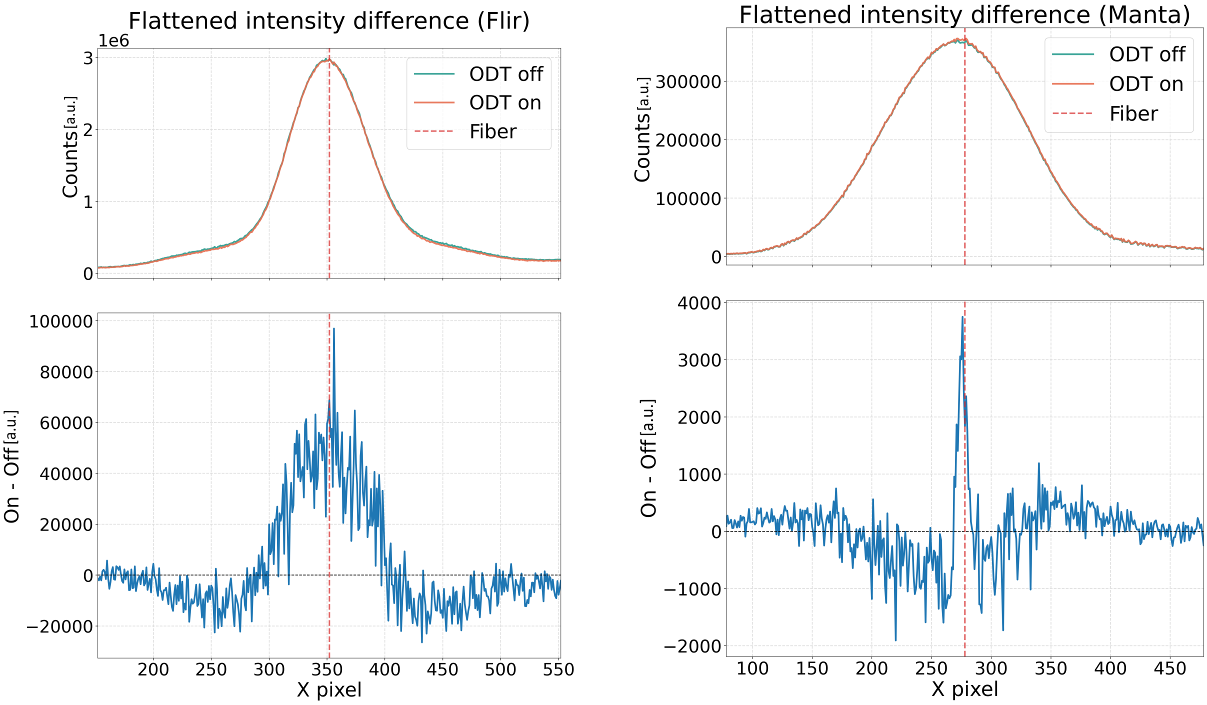
$$\frac{dP}{dA} \approx 0. \tag{1.9}$$

where P is the diffracted light power and A is the AOM amplitude.

As shown by the comparison in Figure 1.6, the spatial sampling was improved by replacing the FLIR camera with the Manta camera, which has a smaller pixel pitch. The Manta camera was also positioned closer to the vacuum chamber, near the minimum working distance of the MVL50M23 objective, to increase the optical magnification. From a comparison of the apparent fiber width recorded with the two imaging configurations, the number of pixels spanning the same object was estimated to increase by a factor of approximately 2.25, yielding a pixel size of $C \approx 19\,\mu\text{m/px}$.

With the final configuration, the broad fluorescence distribution of the freely expanding molasses cloud and the narrower contribution of the ODT-trapped atoms could be distinguished in the same integrated profile, as illustrated in Figure 1.6. This separation is essential for the subsequent analysis: the broad component provides the reference signal for the initial cloud, whereas the narrow differential component determines the ODT signal area, the loading efficiency, and the absolute number of trapped atoms according to the procedure described in Section 2.2.5 and in the corresponding atom-number calibration section.

The final detection configuration, consisting of a 5 ms frequency ramp to the optimal value derived in Section 1.1, and a 5 ms AOM ramp to an AOM amplitude of 480 a.u.($\approx 6.6 mW$), and the Manta imaging system, was used for all subsequent ODT-loading measurements.



(a) *FLIR camera* (b) *Manta camera*

Figure 1.6: *Comparison of the one-dimensional differential fluorescence profiles acquired with the initial FLIR configuration (a) and the upgraded Manta imaging system (b). The upgraded system and data analysis provide better spatial resolution and reduced background fluctuations.*

1.2.2 Capture optimization

The compensation coil currents were optimized using the complete analysis procedure introduced in Section 2.2.5. For every current setting, three related quantities were evaluated: the integrated ODT fluorescence signal, the loading efficiency, and the calibrated number of trapped atoms. The ODT signal area provides a direct measure of the trapped fluorescence, whereas the loading efficiency normalizes this signal to the size of the initial atomic sample. The atom-number calibration converts the integrated counts into an absolute estimate of the ODT population.

The three compensation currents were scanned sequentially in the order

$$x \to z \to y. \tag{1.10}$$

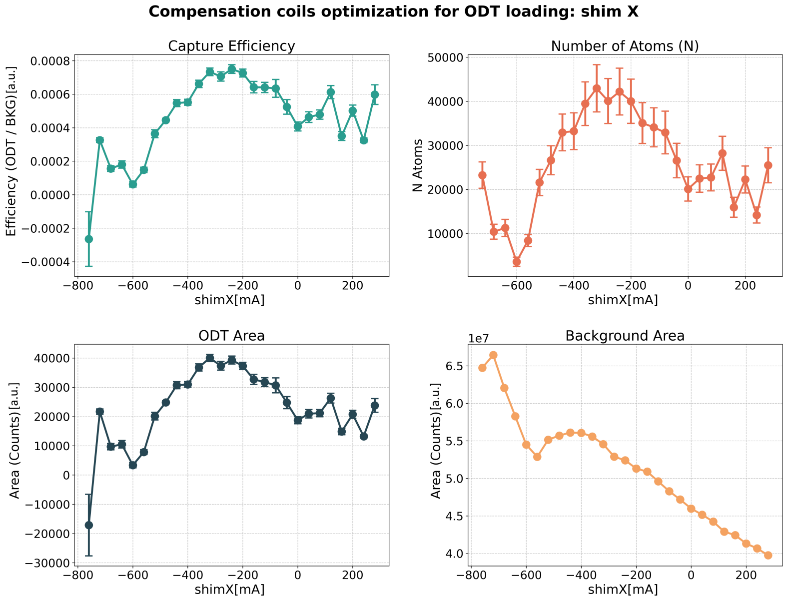
During each scan, the currents along the axes not yet optimized were initially kept at the preliminary values given in equation (1.7). Once an optimum was identified, the corresponding current was retained during the following scans. The integrated ODT signal, the loading efficiency, and the number of trapped atoms exhibit consistent trends as functions of the compensation currents. This agreement indicates that the observed maxima reflect an actual improvement in ODT loading rather than a change caused solely by fluctuations in the initial molasses population.

The scans shown in Figures 1.7a, 1.7b, and 1.7c yielded

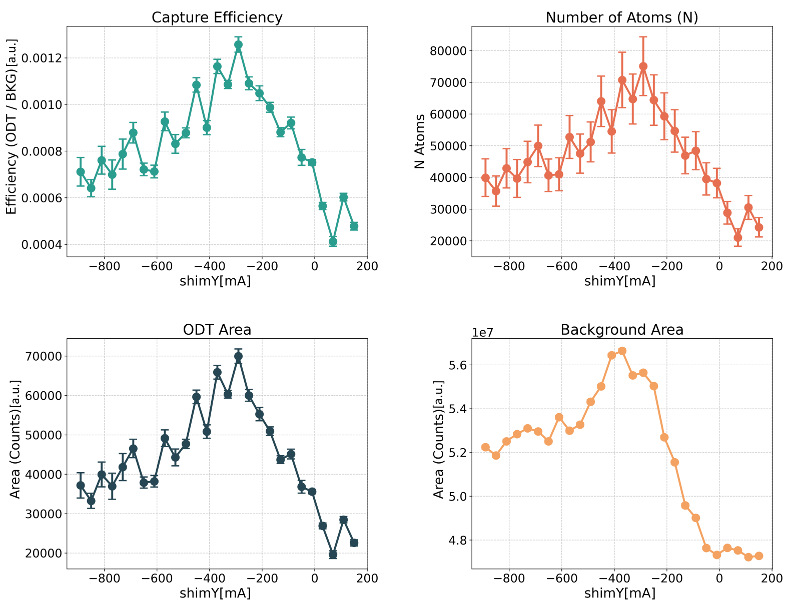
$$\begin{cases} I_x & = -220 \text{ mA}, \\ I_y & = -290 \text{ mA}, \\ I_z & = -90 \text{ mA}. \end{cases} \tag{1.11}$$

A second series of measurements was subsequently performed to verify the reproducibility of the optimized configuration. The scan of the z-axis compensation current produced a more complex behavior than expected. Rather than displaying a single smooth maximum, both the ODT signal and the loading efficiency varied non-monotonically over the investigated range. Starting from $I_z = -1000$ mA, the signal increased toward a local maximum near -800 mA, decreased around -700 mA, and then exhibited additional maxima near -600 mA and -400 mA. The final results of the optimization sequence are summarized in Figure 1.7.

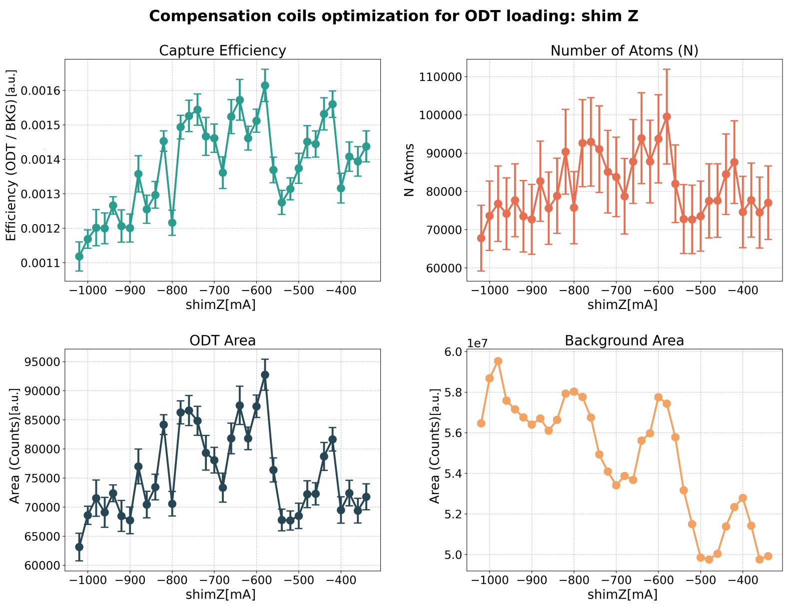
A similar modulation was observed in the broad molasses contribution measured after the time of flight with the ODT switched off. Its maxima occurred near -800 mA, -600 mA, and -400 mA, approximately matching those of the trapped signal. The cloud centroid projected onto the camera x axis also exhibited a correlated displacement, which was not observed during the scans of the other two compensation axes.



(a) *Shim X*



(b) *Shim Y*



(c) *Shim Z*

These observations indicate that changing the z-axis compensation current does not produce only a simple translation of the cloud toward or away from the ODT. It also modifies the position of the MOT within the three-dimensional intensity distribution of the cooling beams. Residual beam misalignments or intensity gradients may therefore cause non-monotonic changes in the MOT population, temperature, or trajectory. The present measurements do not allow a conclusive identification of the origin of this behavior, but they show that both the initial cloud population and the loading fraction must be monitored when selecting the optimal current.

For this reason, the final operating point was chosen by considering the loading efficiency together with the integrated ODT signal and the corresponding calibrated atom number. The selected compensation currents were

$$\begin{cases} I_x &= -220 \text{ mA}, \\ I_y &= -290 \text{ mA}, \\ I_z &= -600 \text{ mA}. \end{cases} \qquad \begin{cases} B_x &= -308 \text{ mG}, \\ B_y &= -377 \text{ mG}, \\ B_z &= -900 \text{ mG}. \end{cases} \tag{1.12}$$

The corresponding displacement of the MOT is estimated from

$$t_\alpha = \frac{B_\alpha}{B'} \qquad (\alpha = x, y), \qquad t_z = -\frac{B_z}{2B'}, \tag{1.13}$$

where $2B' = 9.32$ G/cm is the axial gradient of the quadrupole field. This gives

$$\begin{cases} t_x &= -661\ \mu\text{m}, \\ t_y &= -809\ \mu\text{m}, \\ t_z &= 966\ \mu\text{m}. \end{cases} \tag{1.14}$$

The selected configuration maximized the trapped-atom signal. Its one-dimensional differential profile is shown in Figure 1.8, while Figure 1.9 shows the corresponding two-dimensional differential image. This configuration yielded a loading efficiency of

$$\eta = (0.161 \pm 0.005)\%. \tag{1.15}$$

The fluorescence calibration yields a corresponding ODT population of

$$N_{\text{ODT}} = (1.0 \pm 0.1) \times 10^5 \text{ atoms}. \tag{1.16}$$

which corresponds to an initial molasses population measured in this displaced configuration ($N_{\text{MOT}} \approx 6.2 \times 10^7$ atoms). This value is a lower estimate because the fluorescence calibration approximates the atom as a two-level system: the actual multilevel structure introduces a multiplicative correction factor in the relation between detected fluorescence and atom number [2]; this factor was not included in the value reported in equation (1.16).

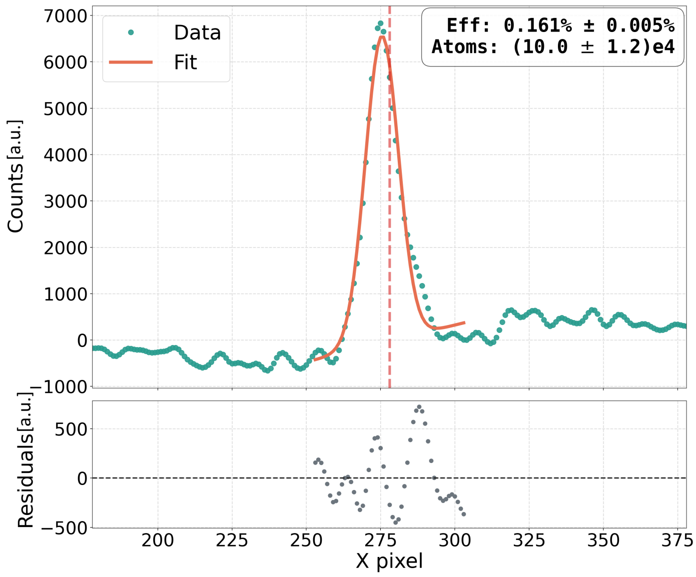


Figure 1.8: *One-dimensional differential fluorescence profile associated with the atoms captured in the optical dipole trap. The narrow component is obtained after background subtraction.*

Figure 1.10 shows the two-dimensional differential fluorescence images acquired at different times of flight and documents the temporal evolution of the cloud released from the ODT, revealing that the trapped atoms remain confined(most intense stripe), while the surrounding atoms are progressively lost.

1.2.3 Comparison with the capture model

The measured loading efficiency can be compared with the estimate obtained from the shallow capture model introduced in Section 2.3. Using the parameters associated with the optimized molasses configuration,

$$w_0 = 19.0\ \mu\text{m}, \qquad P = (0.59 \pm 0.06)\ \text{W}, \tag{1.17}$$

$$R_x R_y = (1.043 \pm 0.004)\ \text{mm}^2, \qquad T = (8.42 \pm 0.17)\ \mu\text{K}, \tag{1.18}$$

the model predicts

$$\eta_{\text{shallow}} = (0.51 \pm 0.06)\%. \tag{1.19}$$

Using the initial number of atoms measured in the displaced MOT, $N_{\text{MOT}} \approx 6.2 \times 10^7$, the theoretically expected number of loaded atoms according to the shallow-trap model is:

$$N_{\text{ODT}}^{\text{theory}} = \eta_{\text{shallow}}\, N_{\text{MOT}} \approx (3.2 \pm 0.4) \times 10^5\ \text{atoms}. \tag{1.20}$$

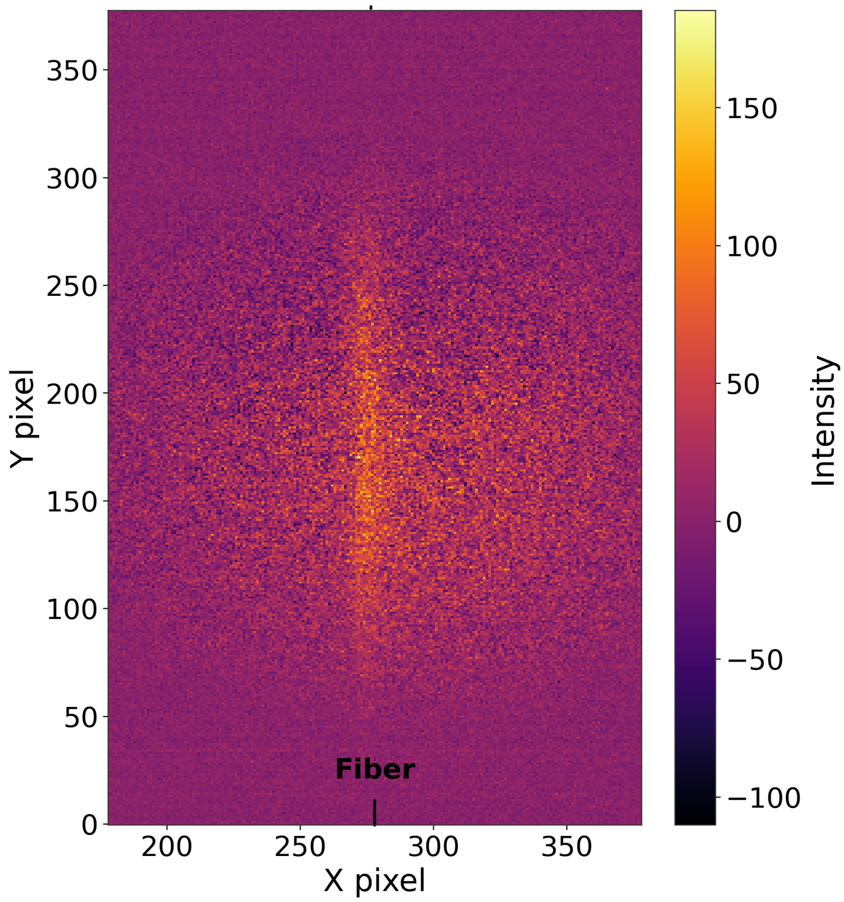
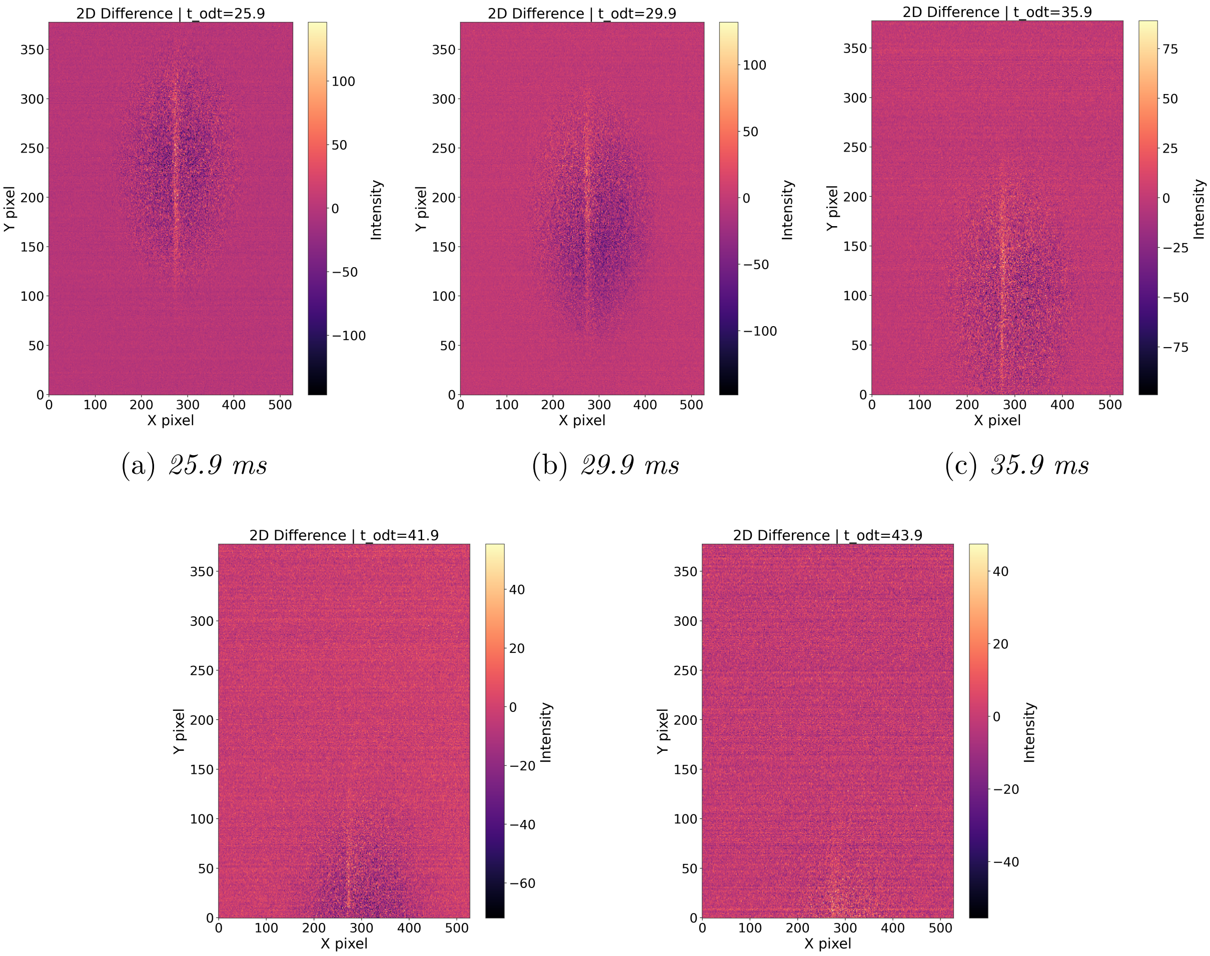


Figure 1.9: *Two-dimensional differential fluorescence image acquired with the selected compensation currents,* $I_x = -220$ mA*,* $I_y = -290$ mA*, and* $I_z = -600$ mA*. This configuration gives a loading efficiency of* $\eta = (0.161 \pm 0.005)\%$ *and corresponds to the largest calibrated number of ODT-trapped atoms.*

In Section 2.3, the expression used to estimate the number of atoms guided toward the fiber includes a factor of 1/2, accounting for atoms with $v_z < 0$. This factor is omitted here because the present measurement concerns the total population captured by the ODT, irrespective of the sign of the axial velocity.

The measured efficiency and trapped atom number are of the same order of magnitude as the theoretical estimate, indicating that the selected operating conditions successfully load the dipole trap. However, the model predicts a capture efficiency and an expected atom number approximately 3.2 times larger than the experimentally measured values. Several effects may contribute to this difference. In particular, the calculation assumes a Gaussian ODT profile, whereas the field transmitted through the fiber contains higher-order modes in addition to the fundamental LP_{01} mode. Spatial distortions of the trapping potential may consequently reduce the effective capture volume relative to the idealized model.

Possible losses introduced by the measurement sequence must also be considered. In the present procedure, the ODT is switched off for 600 μs during the final fluorescence acquisition. Atoms may leave the detection region during this interval, causing the measured



(a) *25.9 ms* (b) *29.9 ms* (c) *35.9 ms* (d) *41.9 ms* (e) *43.9 ms*

Figure 1.10: *Two-dimensional differential fluorescence images showing the temporal evolution of the atomic cloud trapped in the ODT at different times of flight.*

loading efficiency and atom number to underestimate the populations initially captured by the trap. The experimental results should therefore be interpreted as conservative estimates of the ODT loading performance. This interpretation also accounts for the two-level approximation used in the atom-number calibration: because the actual atom has a multilevel structure, the omitted multiplicative correction factor leads to a further underestimate of the trapped atom number, but cancels out in the ratio and therefore does not affect the measured capture efficiency.

1.3 Power and polarization stability results

The black-fiber injection system described in Chapter 2 provides the counterpropagating optical field required to generate the conveyor-belt potential. For this application, stable optical power and polarization are essential. Power fluctuations directly affect the lattice depth and may lead to parametric heating, whereas polarization fluctuations may reduce the interference contrast and, consequently, the reproducibility of the optical potential.

This section presents the experimental characterization of the power and polarization stability of the optical field transmitted through the black fiber. The system is analyzed both in the time domain, using the fully overlapping Allan deviation, and in the frequency domain, using the Amplitude Spectral Density (ASD). These complementary analyses make it possible to identify the relevant noise processes, determine the timescales over which the system is most stable, and compare the measured noise with the theoretical shot-noise limit.

1.3.1 Time-domain stability

The time-domain stability of the transmitted optical power and polarization was evaluated from a continuous acquisition lasting 600 s. Figure 1.11 displays the raw voltage signals for the power channels and the corresponding fluctuation of the polarization angle recorded during this time window. The fully overlapping Allan deviation was then calculated for the recorded signals in order to distinguish short-term fluctuations from slow environmental drifts.

The top panel of Figure 1.12 shows the fractional Allan deviation $\sigma_y(\tau)$ of the optical power. The total power, obtained as the sum of the transmitted and reflected channels, reaches a minimum fractional Allan deviation of approximately 8×10^{-4} at an averaging time of $\tau \approx 0.1$ s. For shorter averaging times, $\sigma_y(\tau)$ follows the characteristic $1/\sqrt{\tau}$ dependence associated with white noise. For $\tau > 0.1$ s, the Allan deviation increases progressively, exceeding 10^{-2} at $\tau = 200$ s. This behavior indicates that the long-term power stability is limited by low-frequency processes, most likely associated with thermal drifts and slow mechanical relaxation of the optical components.

The bottom panel of Figure 1.12 shows the angular Allan deviation $\sigma_\theta(\tau)$ of the polarization angle. The minimum value, approximately 7×10^{-5} rad, is obtained at an averaging time of $\tau \approx 3$ s. At shorter averaging times, the Allan deviation follows the $1/\sqrt{\tau}$ dependence; for $\tau > 3$ s, slow environmental drifts become dominant and $\sigma_\theta(\tau)$ increases, reaching approximately 8×10^{-4} rad at $\tau = 200$ s.

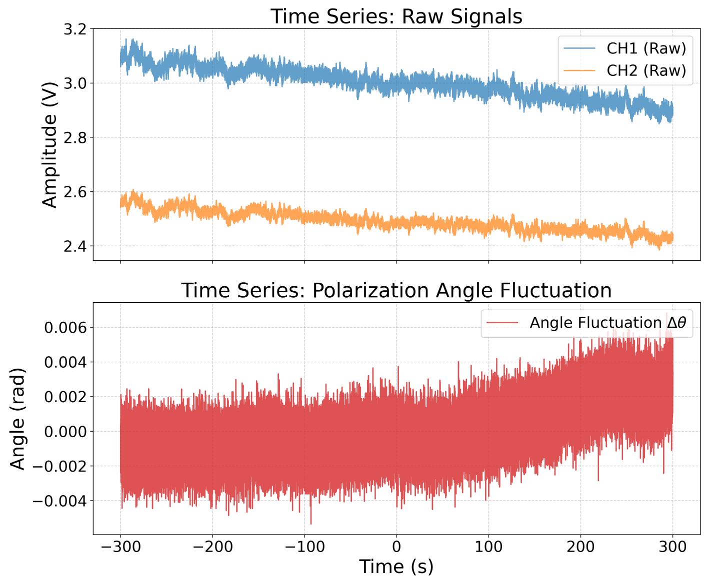


Figure 1.11: *Time series of the raw signals measured after the black fiber during a* 600 s *acquisition. Top: voltage amplitude of the two transmission channels (CH1 and CH2). Bottom: corresponding fluctuation of the polarization angle* $\Delta\theta$.

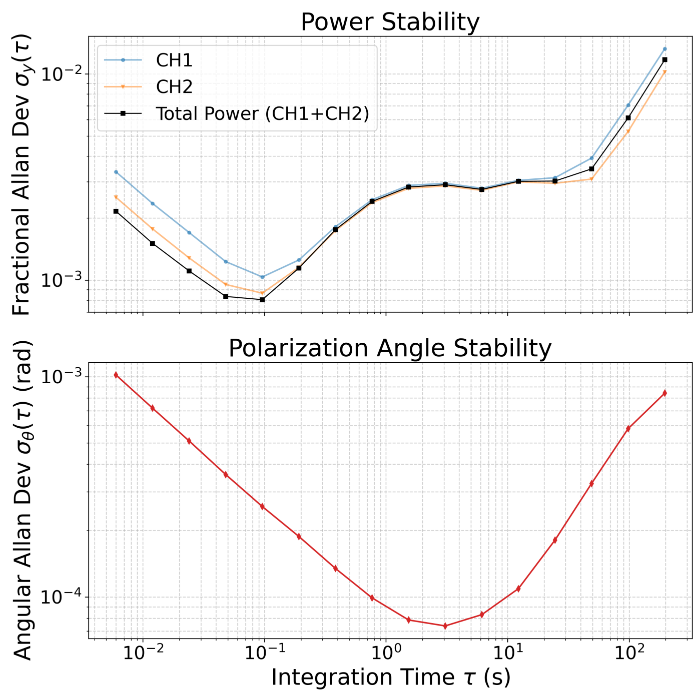


Figure 1.12: *Fully overlapping Allan deviation of the power and polarization signals measured after the black fiber over a* 600 s *acquisition. Top: fractional Allan deviation* $\sigma_y(\tau)$ *of the individual power channels and their sum. The total power reaches a minimum fractional Allan deviation of approximately* 8×10^{-4} *at* $\tau \approx 0.1$ s. *Bottom: angular Allan deviation* $\sigma_\theta(\tau)$ *of the polarization angle, with a minimum of approximately* 7×10^{-5} rad *at* $\tau \approx 3$ s. *At longer averaging times, both quantities are increasingly affected by slow environmental drifts.*

1.3.2 Frequency-domain analysis

The Allan-deviation analysis identifies the timescales over which the transmitted field is most stable, but it does not directly reveal the spectral distribution of the underlying fluctuations. The acquired signals were therefore also analyzed in the frequency domain by calculating the Amplitude Spectral Density (ASD) of the photocurrent. The measured spectra were compared with the theoretical shot-noise levels calculated using the photocurrents recorded in the corresponding detection channels.

Figure 1.13 shows the low-frequency spectra obtained from the 600 s acquisition over the range from 10^{-3} to 10^{2} Hz. Below approximately 1 Hz, the spectra are dominated by excess technical noise and exhibit a clear low-frequency increase consistent with the long-term drifts observed in the Allan deviation. As the frequency increases, the measured noise decreases, although it remains significantly above the calculated white-noise floors, represented by the dashed lines, clearly showing the presence of technical noise.

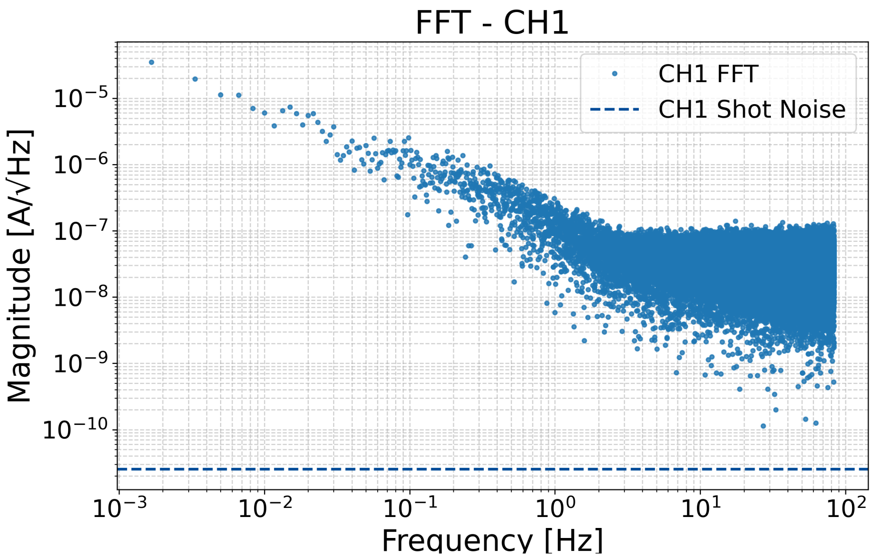


Figure 1.13: *Amplitude Spectral Density of the photocurrent fluctuations measured after the black fiber over the frequency range from* 10^{-3} *to* 10^{2} Hz. *The dashed lines represent the calculated white-noise floors for the corresponding detection channels. Excess technical noise dominates the spectra below approximately* 1 Hz.

An additional measurement based on a shorter 600 ms acquisition with a larger bandwidth was performed to characterize the system at frequencies up to 10^{5} Hz. The resulting transmitted-photocurrent spectrum is shown in Figure 1.14. Residual technical noise is visible below approximately 10^{3} Hz. The spectrum flattens and becomes comparable to the calculated shot-noise level of 10^{-11} A/$\sqrt{\text{Hz}}$. This result demonstrates that the fiber does not introduce any intrinsic limitations, allowing the system to reach the fundamental detection noise floor at higher frequencies.

The time- and frequency-domain measurements provide a consistent characterization of

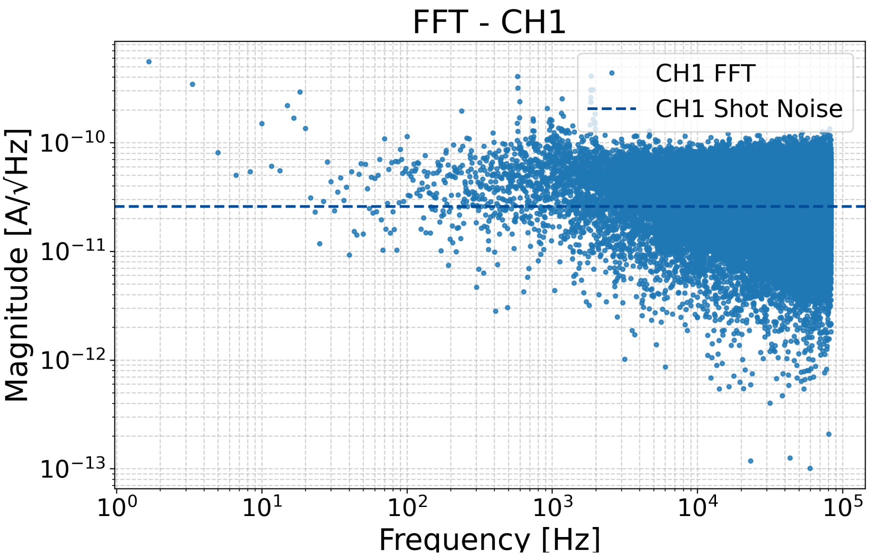


Figure 1.14: *Amplitude Spectral Density of the transmitted photocurrent up to* 10^5 Hz. *The red dashed line represents the calculated photocurrent shot-noise level of* 10^{-11} A/$\sqrt{\text{Hz}}$. *The measured spectrum is comparable with the shot-noise floor.*

the stability of the optical field delivered through the black fiber. Two estimates relate these measurements to the requirements of the ^{87}Rb optical conveyor belt.

Intensity noise heats trapped atoms at a rate $\Gamma = \pi^2\nu^2 S_I(2\nu)$, where ν is the trap frequency and S_I is the intensity noise spectrum [3, 4]. Optical conveyor belts operate with trap frequencies of order 10–400 kHz, so the relevant noise frequency 2ν falls in the range where the measured ASD is already at the shot-noise floor (Fig. 1.14). At the shot-noise limit, the heating rate is of order $\Gamma \sim 10^{-6}$ s^{-1}, corresponding to a heating time of $\sim 10^6$ s, well above any relevant experimental timescale.

The top-down and bottom-up beams of the conveyor belt are delivered through two separate fibers, the black fiber and the HCPCF. A fluctuation of the polarization angle measured after the black fiber therefore contributes directly to a fluctuation of the relative angle $\delta\theta$ between the two interfering polarizations, and hence to the contrast of the standing wave. At $\tau = 200$ s, the measured angular Allan deviation reaches approximately $\sigma_\theta \simeq 8 \times 10^{-4}$ rad. Using this value as an order-of-magnitude estimate for the relative polarization fluctuation, the normalized polarization overlap changes as

$$\cos(\delta\theta) \simeq 1 - \frac{\delta\theta^2}{2},$$

corresponding to a relative contrast reduction of approximately 3×10^{-7}. This estimate indicates that the polarization fluctuations of the black-fiber branch alone are unlikely to limit the conveyor-belt contrast. However, the polarization stability of the HCPCF branch was not measured separately, so the stability of the complete two-beam lattice cannot be inferred from this measurement alone.

Both estimates indicate that the noise introduced by the black fiber is compatible with the requirements of the ^{87}Rb optical conveyor belt. The same Allan-deviation and ASD analysis can be repeated after integration of the upper breadboard, providing a diagnostic tool to identify and localize any additional noise sources introduced downstream of the fiber.

1.4 Numerical Simulation Results of EIT Cooling

The numerical model described in Section 2.7 was evaluated to characterize the Electromagnetically Induced Transparency (EIT) cooling dynamics of ^{87}Rb atoms. To distinguish the fundamental cooling mechanism from the effects of the complex atomic structure, the simulations were performed in two stages. First, an idealized three-level Λ system was simulated to validate the analytical predictions. Subsequently, a complete 24-level model incorporating the full hyperfine and Zeeman structure of the D_2 line was implemented to assess realistic experimental conditions, including off-resonant excitation and population leakage.

1.4.1 Ideal three-level system dynamics

The physical parameters used in the idealized three-level simulation, together with the assumed beam propagation geometry, are summarized in Table 1.1.

Table 1.1: *Simulation parameters for the idealized three-level Λ system.*

Parameter	Symbol	Value
Trap frequency	ν	0.5γ
Coupling detuning	Δ_c	15.0γ
Probe detuning (two-photon resonance)	Δ_p	Δ_c
Coupling Rabi frequency	Ω_c	$\sqrt{4\lvert\Delta_c\rvert\nu}$
Probe Rabi frequency	Ω_p	0.1γ
Lamb–Dicke parameter	η	0.35
Coupling beam propagation	–	orthogonal to the trap axis ($\eta_c \simeq 0$)
Probe beam propagation	–	along the trap (motional) axis

Since this idealized treatment collapses the atomic structure into three bare levels without an explicit magnetic sublevel structure, no specific light polarization is associated with either field: the coupling and probe transitions are simply assumed to be independently addressable.

The steady-state spectroscopic response of the ideal three-level system was obtained by scanning the probe-laser detuning Δ_p while keeping the coupling field fixed. The resulting total excited-state population exhibits the characteristic Fano profile associated with EIT. Figure 1.15 shows both the macroscopic transparency window and a detailed view of the sideband alignment.

The coupling laser intensity was set according to the approximate cooling condition (Eq. (2.99)), tuning the AC Stark shift so that the absorption maximum overlaps with the cooling transition. The time evolution of the average phonon number $\langle n\rangle$ for this

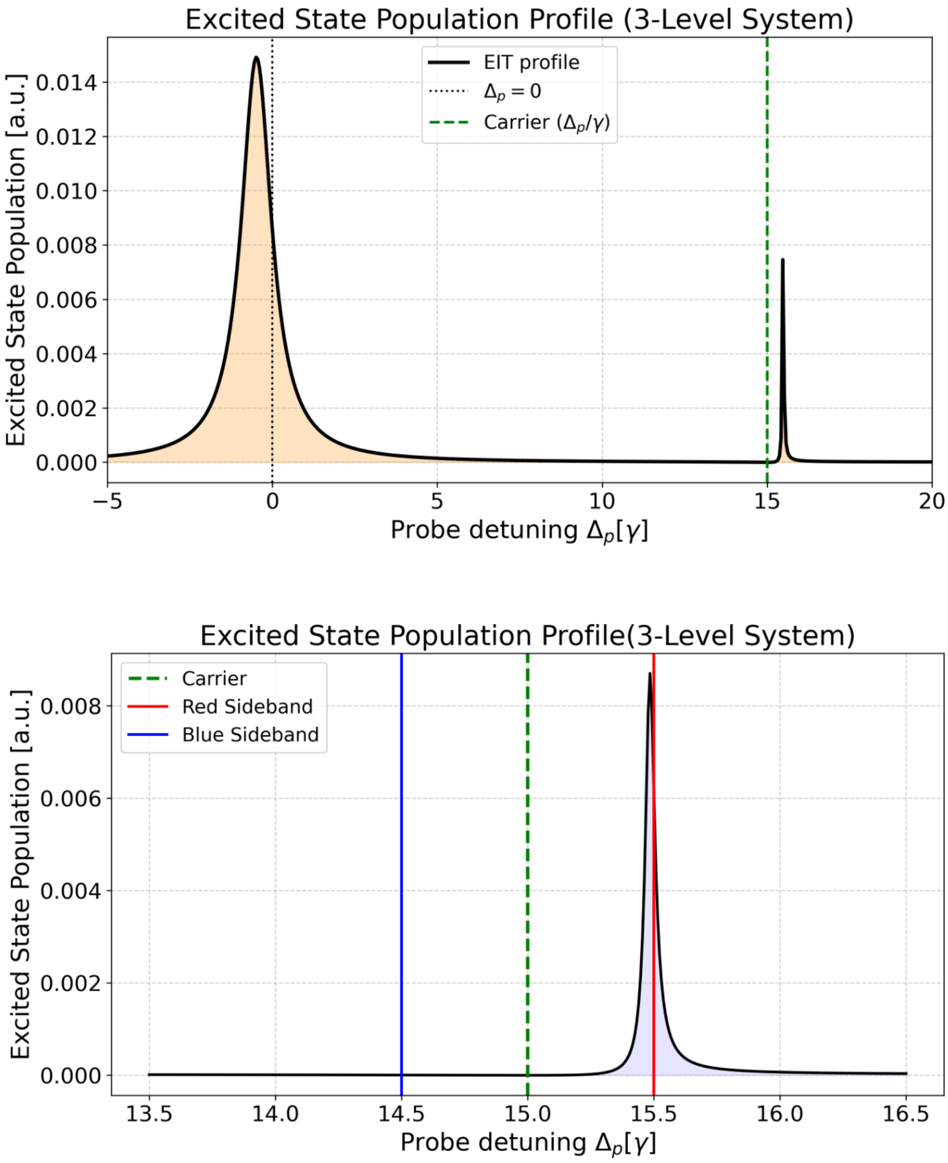


Figure 1.15: *Steady-state excited population profile of the idealized three-level system. Top: Full spectrum showing the transparency window (dashed green line) at the two-photon resonance. Bottom: Detailed profile near the dark state, showing the narrow absorption maximum aligned with the red sideband (solid red line) and the carrier transition falling in the transparency minimum (dashed green line).*

configuration is presented in Figure 1.16. Starting from a Fock state with $n = 15$, the system undergoes rapid ground-state cooling, decaying exponentially and approaching the theoretical minimum limit in less than 0.20 ms. This confirms the efficiency of the targeted sideband asymmetry in the absence of competing decay channels.

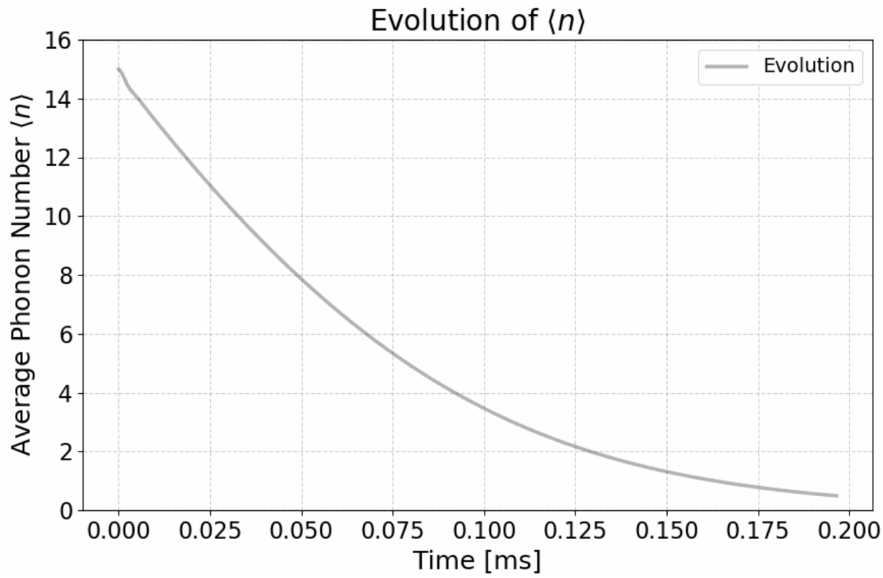


Figure 1.16: *Time evolution of the average phonon number $\langle n \rangle$ in the three-level simulation. The population rapidly cools to the motional ground state within approximately 0.20 ms.*

1.4.2 Multilevel simulation and optical pumping effects

Extending the simulation to the full 24-level structure of the ^{87}Rb D_2 line introduces significant complexity due to spontaneous emission into ground states outside the designed Λ subsystem. The simulation parameters were optimized for the axial conveyor belt geometry and are summarized in Table 1.2, and N_vib is set to 10.

To demonstrate the necessity of repopulating the active cooling cycle via the repump field, the dynamics were first simulated without an active repump field, and then compared with the full optimized sequence. The impact of optical pumping is evident in the steady-state spectroscopy (Figure 1.17). Without a repump field, the excited-state population drops to near zero across the entire frequency scan. Following a few excitation cycles, the atoms decay into uncoupled Zeeman sublevels, depleting the active EIT levels.

By introducing a π-polarized repump field, the atoms are recycled back into the EIT cycle. The Fano profile and the dark resonance are recovered, allowing the alignment of the red sideband with the absorption peak. However, compared to the three-level model, the resonance is broader and the transparency minimum is less deep, owing to off-resonant coupling to adjacent excited manifolds.

Table 1.2: *Optimal simulation parameters for the 24-level EIT cooling model, corresponding to the axial conveyor belt geometry.*

Parameter	**Symbol**	**Value**
Axial trap frequency	ν_z	430 kHz
Lamb–Dicke parameter	η	0.09
Initial mean phonon number	$\langle n_0 \rangle$	2
Magnetic field	B	4.0 G
Probe Rabi frequency	Ω_p	0.09γ
Probe polarization	q_p	σ^-
Probe beam propagation	–	along the trap (axial) direction
Coupling detuning	Δ_c	$+13\gamma$
Coupling Rabi frequency	Ω_c	$\sqrt{4\nu_z(\nu_z + \lvert\Delta_c\rvert)}$
Coupling polarization	q_c	σ^-
Coupling beam propagation	–	orthogonal to the axial direction
Repump Rabi frequency	Ω_r	0.3γ
Repump polarization	q_r	π
Repump beam propagation	–	orthogonal to the axial direction

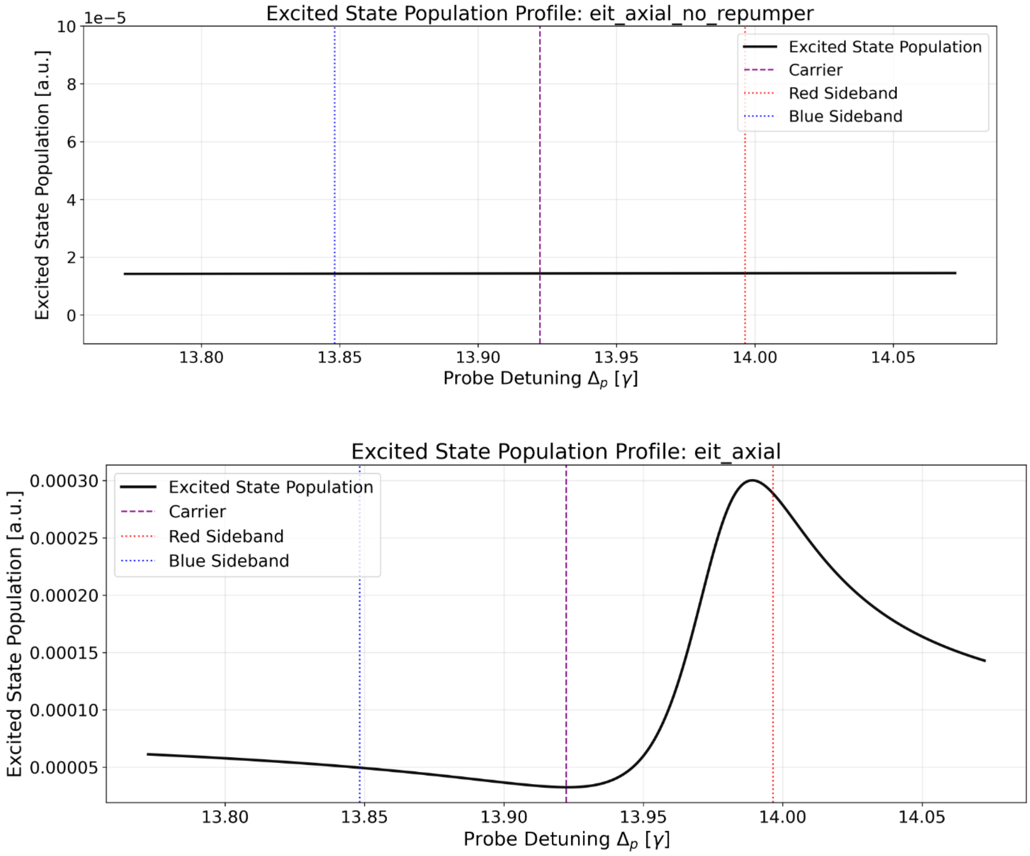


Figure 1.17: *Comparison of the steady-state excited population in the 24-level system. Top: Without a repump field, optical pumping into dark states causes the spectroscopic signal to vanish. Bottom: With an active repump field, the Fano profile is restored, though broadened by off-resonant scattering.*

This optical pumping mechanism is directly observable in the time evolution of the internal atomic populations (Figure 1.18). Without the repumper, the active ground state populations decay rapidly, accumulating entirely in the uncoupled states within a few milliseconds. Conversely, activating the repump field sustains the population within the active states throughout the evolution, continuously recovering atoms that leak out of the ideal cooling cycle.

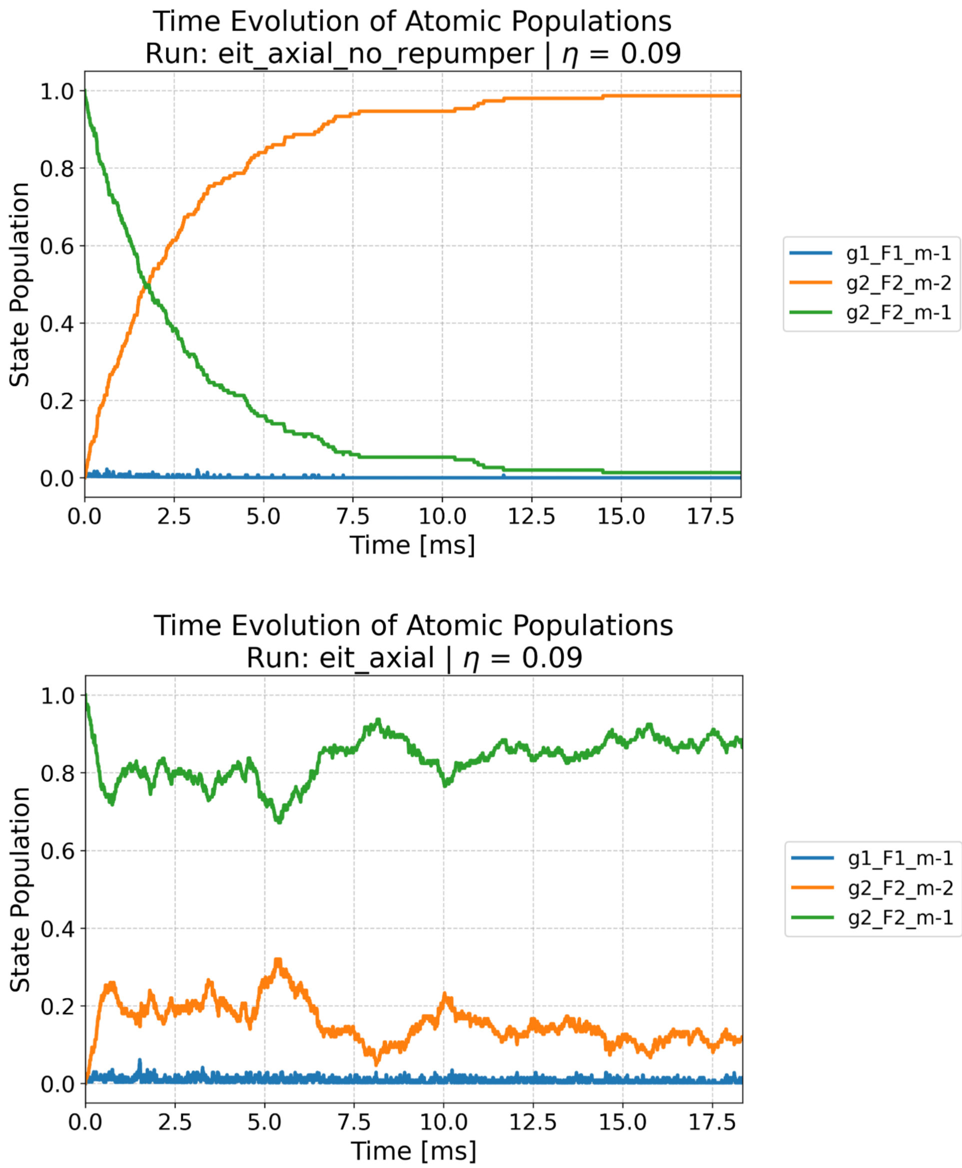


Figure 1.18: *Time evolution of the atomic populations. Top: Without a repumper, atoms rapidly leave the active EIT cycle and accumulate in uncoupled dark states. Bottom: The continuous action of the repump field stabilizes the active populations, allowing the cooling cycle to persist.*

The consequence of this complex internal structure on the final motional state is evaluated using Monte Carlo trajectory simulations (`mcsolve`). Figure 1.19 compares the cooling dynamics with and without the repump field. When optical pumping depletes the active states, the cooling mechanism is interrupted: the phonon number initially decreases but plateaus around $\langle n \rangle \approx 1.1$, preventing ground-state cooling. With the repump field active, EIT sideband cooling is sustained, ultimately bringing the atoms toward the motional ground state ($\langle n \rangle \ll 1$), reaching a final value of $\langle n \rangle \approx 0.08$. A direct quantitative comparison with the analytical steady-state prediction is not straightforward, however, since the latter would require the photon recoil to be explicitly included in the theoretical treatment.

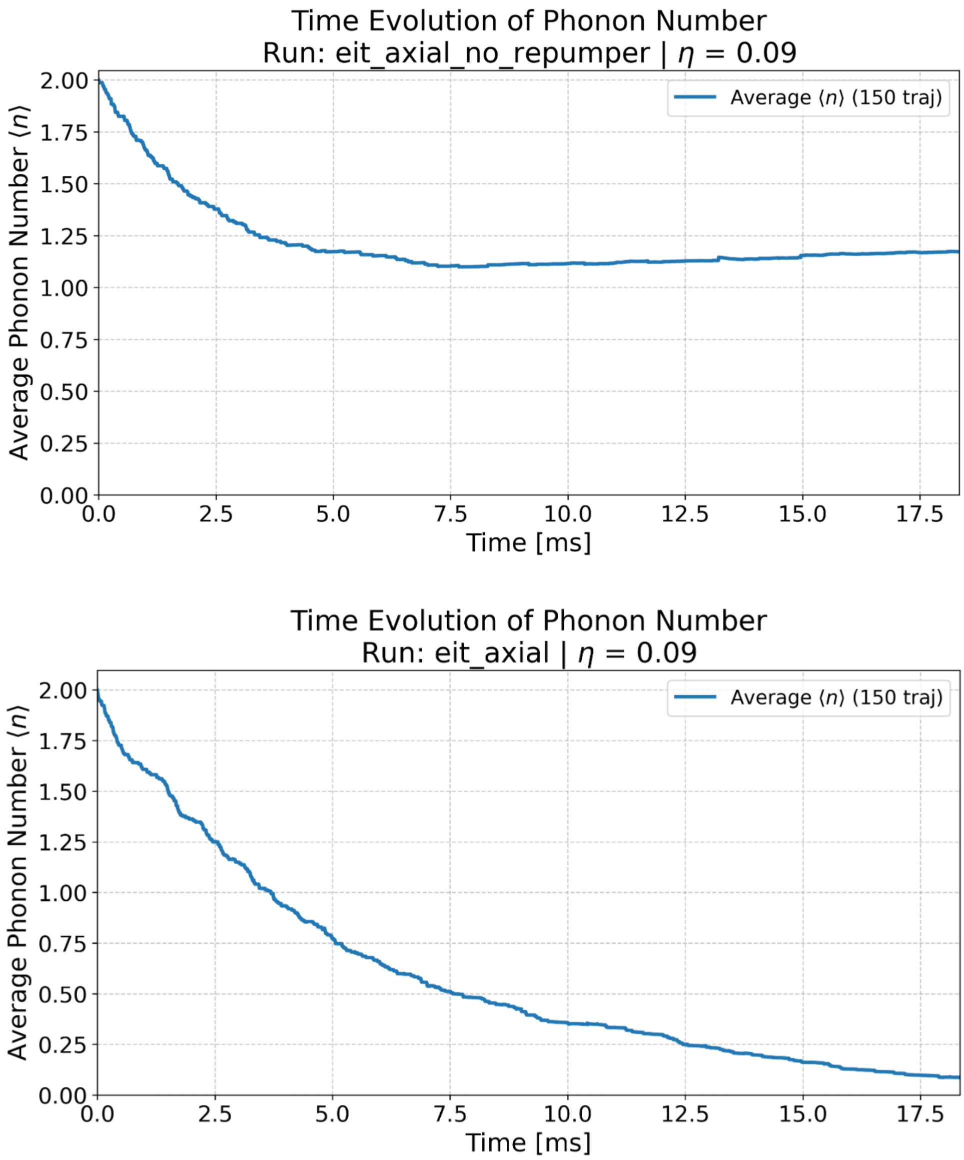


Figure 1.19: *Time evolution of the average phonon number $\langle n \rangle$ in the 24-level system (averages of 150 quantum trajectories). Top: Without a repump field, cooling halts as atoms become trapped in uncoupled states. Bottom: With the repump field, cooling to $\langle n \rangle \ll 1$ is successfully achieved over roughly 18 ms.*

However, the effective cooling rate is drastically reduced compared to the ideal three-level case. The system requires nearly 18 ms to approach $\langle n \rangle \approx 0$, two orders of magnitude longer than the idealized simulation. First, atoms spend a significant fraction of their time outside the optimal Λ system waiting to be recycled; second, the imperfect carrier suppression introduce heating events that must be continuously compensated. Despite this reduction in cooling speed, the simulated cooling timescale is compatible with state-of-the-art EIT cooling experiments implemented in free-space setups, where similar multilevel effects occur and a 20 ms cooling time is achieved [5].

While these results validate the EIT cooling scheme within the complete atomic level structure, the simulation must eventually be adapted to reflect the exact conditions of the fiber implementation. Atoms confined near the core surface of the hollow-core fiber will experience spatially varying tensor light shifts, local magnetic field gradients, and specific polarization constraints due to the fiber's geometry. Future modeling efforts will need to incorporate these physical perturbations into the Hamiltonian to fully optimize the laser parameters and ensure robust cooling in the final experimental environment.

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